

- 13** A gas is stored in a container of constant volume.

The temperature of the gas is increased.

What happens to the particles of the gas?

- A** The distance between the particles increases.
- B** The mass of the particles increases.
- C** The number of particles increases.
- D** The speed of the particles increases.

- 14** One end of a shiny metal rod is heated, and the other end quickly gets hot.

Which statement describes why the other end quickly gets hot?

- A** Metals are good thermal conductors.
- B** Metals are poor thermal conductors.
- C** Shiny surfaces are good emitters of infrared radiation.
- D** Shiny surfaces are poor emitters of infrared radiation.

- 15** Vacuum flasks usually have silvered walls that help to keep the contents of the flask hot.

Why are the walls silvered?

- A** to absorb thermal energy from the air around the flask
- B** to increase the rate of convection inside the flask
- C** to reduce energy loss to the surroundings by conduction
- D** to reflect thermal radiation back into the flask

- 16** The temperature of the air in the atmosphere around us increases.

Which row describes the change in the density of the air and which form of thermal energy transfer occurs as a result of this change?

	what happens to the density of the air	thermal energy transfer that occurs as a result
A	decreases	conduction
B	decreases	convection
C	increases	conduction
D	increases	convection